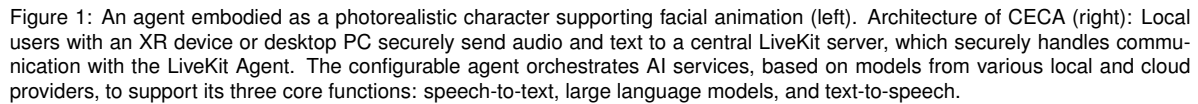


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We present CECA, a configurable framework for embodied conversational AI agents in Unity-based extended reality (XR) applications. CECA employs a client-server architecture to decouple agent logic from game engine-based embodiment. Built on LiveKit Agents, our approach integrates speech-to-text (STT), large language models (LLMs), and text-to-speech (TTS) into a unified, streaming voice-to-voice pipeline configured via metadata rather than code changes. We outline how this architecture flexibly integrates local and cloud AI providers while mitigating limited provider SDK support in Unity. Finally, we highlight opportunities for future work, including multi-agent scenarios, higher-level templates for XR research, and systematic user studies.

Embodied conversational agents (ECAs) have been studied for several decades, with early systems relying on pre-scripted dialog and behavior controllers such as finite-state machines or rule-based approaches [2]. Recent advances in large language models (LLMs), multimodal perception, and neural speech synthesis now enable more autonomous, adaptive, and lifelike ECAs. Despite this progress, integrating modern conversational agents into extended reality (XR) applications remains a challenging task.

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¹<https://github.com/HardCodeDev777/UnityNeuroSpeech>

2 CECA FRAMEWORK

CECA follows a client-server architecture (see Figure 1). In our reference setup, (1) a LiveKit server runs on an Ubuntu LTS virtual machine and is reachable over the internet, (2) a LiveKit Agent runs on the same machine to reduce end-to-end latency, and (3) the XR application is implemented in Unity and connects using our LiveKit Unity packages. Additionally, we provide a lightweight LiveKit Server SDK for Unity that abstracts room management and agent dispatching. This setup enables the AI logic to be hosted on different machines or platforms than the user's XR client.

2.1 LiveKit Server

LiveKit² is an open-source framework for real-time audio, video, and data communication. At its core is a WebRTC-based Selective Forwarding Unit (SFU), the LiveKit server, which manages transport, routing, and synchronization for multiple participants (e.g., Unity clients and AI agents). The server offers several advantages: (1) hosted on a local PC, a self-hosted server, or managed via the LiveKit Cloud, (2) built-in end-to-end encryption (E2EE), (3) room, participant, and track management, and (4) low-latency audio/video/data streaming. In CECA, the LiveKit server securely handles media transport between Unity clients and LiveKit agents.

2.2 LiveKit Agents

The LiveKit Agents³ framework is also open source and actively developed and supported by the community. It provides higher-level building blocks for implementing AI agents on top of the LiveKit server. It offers AI service integration with a wide range of cloud providers (e.g., Anthropic, Deepgram, ElevenLabs, Gemini, Groq, OpenAI) and also supports local providers (e.g., Kokoro, Olama, Whisper), with new plugins added continually. The LiveKit Agent SDK solves core challenges in real-time voice AI, including streaming audio through an STT-LLM-TTS conversational pipeline, robust turn-taking, event and interruption handling, and coordinating multiple LLM calls, including tooling.

In CECA, the conversational pipeline is implemented as a configurable LiveKit Agent. The agent joins a LiveKit room as a participant, subscribes to the user's audio track, performs continuous speech recognition with turn-taking, generates a response with an LLM, and then produces synthesized speech as an outgoing audio stream. The individual AI services (STT, LLM, TTS) are modular and can be flexibly orchestrated. The agent's configuration is provided at runtime as JSON metadata from the Unity client.

2.3 Unity Integration

On the Unity client side, CECA uses the LiveKit Unity SDK⁴ to connect the XR application to the LiveKit server, and a lightweight LiveKit Server SDK to create and manage rooms, and dispatch agents. Additionally, it utilizes Reality Stack I/O [1] for XR cross-device and cross-platform integration. We provide a thin wrapper for the LiveKit Unity SDK that exposes a single, reusable configuration layer for configuring the agent via JSON-based metadata, including STT, LLM, and TTS service providers, models, and other parameters. This keeps the Unity C# code independent of individual provider SDKs and allows the same integration layer to be reused across agent configurations, with agents running on a local PC, a self-hosted server, or in the LiveKit Cloud.

In CECA, the Unity application joins a LiveKit room, publishes the user's microphone stream, subscribes to the agent's audio track, and renders it as a spatialized audio source in the scene. The audio

source is then connected to a character in the scene, and facial animations are added using the SALSA LipSync Suite⁵ Unity asset. RSIO manages XR device integration in the background, enabling the same CECA setup to be used across different headsets.

3 USE CASES AND PRELIMINARY EVALUATION

To assess the applicability and practicality of CECA, we implemented an example XR application in Unity. This project demonstrates how to configure and orchestrate multiple AI services for a voice-to-voice CECA. The agent is embodied with different character types, including photorealistic characters captured with our photogrammetry scanner, as well as characters from Microsoft Rocketbox and Reallusion Character Creator 4 (CC4). This project serves as an internal prototype and initial validation to demonstrate how heterogeneous AI services and embodiments can be combined via a single LiveKit-based infrastructure.

Researchers in our group reported that CECA reduced integration overhead and made it straightforward to experiment with different AI services, providers, models, prompts, and tools. They emphasized the value of easily exploring and comparing local and cloud providers to find an appropriate fit for their experimental requirements, with respect to latency, robustness, and privacy.

4 DISCUSSION AND FUTURE WORK

This poster demonstrates a novel approach for integrating configurable embodied conversational AI agents into XR applications by combining LiveKit Agents with a metadata-driven configuration layer and Unity-based agent embodiment. While deploying LiveKit adds some initial setup complexity, this is outweighed by the benefits of a purpose-built agent framework that provides advanced real-time voice AI capabilities (e.g., streaming STT-LLM-TTS pipelines, robust turn-taking, interruption handling, and tool use). Decoupling the Unity client from the agent enables AI functionality to evolve independently of the XR application, making it easier to reconfigure or replace components without modifying game engine code. RSIO further abstracts XR device integration, allowing CECA-based agents to be used on different XR devices.

We plan various extensions to CECA, including (1) multi-agent scenarios, (2) action tags for controlling facial animation according to the LLM's suggested voice tone, (3) higher-level templates for common XR research tasks, (4) user studies on social presence, usability, and perceived acceptance, and (5) publishing a reference implementation for developers and researchers.

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⁵<https://crazyminnowstudio.com/unity-3d/lip-sync-salsa/>

²<https://docs.livekit.io/>

³<https://docs.livekit.io/agents/>

⁴<https://github.com/livekit/client-sdk-unity>